

Jum'at 2 Oktober 2020



EDUVISI ONLINE CLASS

BEASISWA

WORKSHEET

Mr. Aris

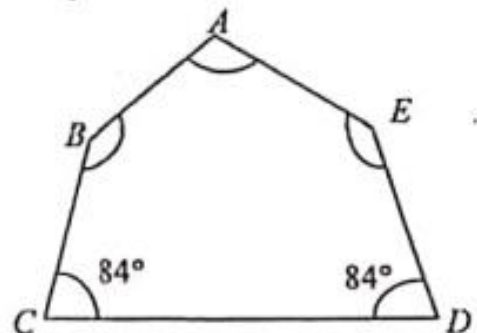
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1. Express $\frac{2x-3}{2x^2+7x-15} - \frac{3}{x+5}$ as a single fraction in its simplest form.

2. In the diagram, $ABCDE$ is a pentagon in which $\angle BCD = \angle CDE = 84^\circ$ and $\angle CBA = \angle BAE = \angle AED$. [Diagram is not drawn to scale.]

(i) Find $\angle BAE$.

(ii) Given that $\angle CBD = 5\angle BDC$, find $\angle CBD$.



3. Given that $a^2 - b^2 = 60$ and $a - b = 6$, find the value of

(i) $a + b$,

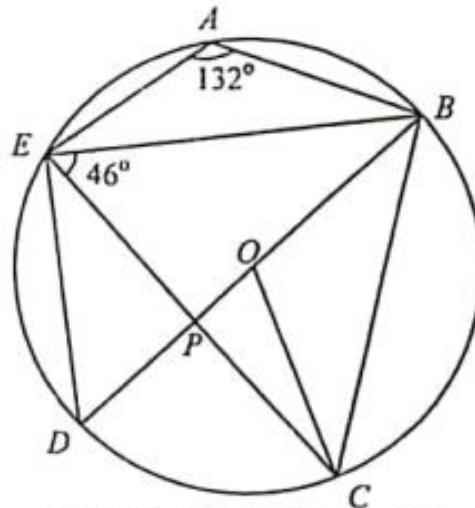
(ii) $a^2 + b^2$.

4. Given that x and y are integers, and $-9 \leq x \leq 8$ and $1 \leq y \leq 7$, find

(i) the smallest possible value of $\frac{x^2}{y}$,

(ii) the largest possible value of $y - 2x$.

5.



In the figure, O is the centre of the circle. DOB is the diameter of the circle, $\angle CEB = 46^\circ$ and $\angle EAB = 132^\circ$. Stating your reasons clearly, calculate

- (i) $\angle BOC$,
- (ii) $\angle DEC$,
- (iii) $\angle EBD$.

6. (i) During the Great Singapore Sale, the price of a Blue-Ray DVD player was reduced from \$240 to \$212. If the player was sold at its original selling price, the shopkeeper would make a profit of 25%. Find the percentage profit when it was sold at the reduced price.

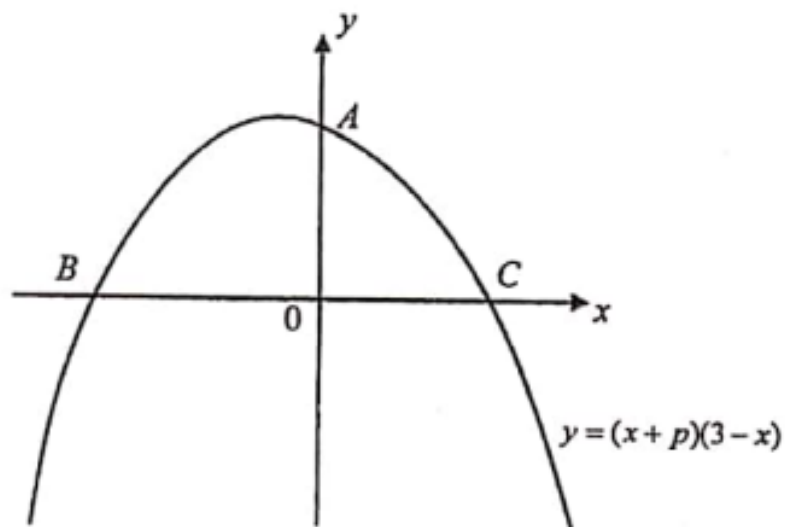
- (ii) The table shows the rate of interest offered by two different banks.

Name of Bank	Interest Rate
POSB	2% per annum compound interest
HSBC	4% per annum simple interest

Mr Tan deposited \$1500 in POSB and \$4500 in HSBC. Calculate the total amount he would have in the banks from both the deposits at the end of two years.

7. (i) A developer estimates that he needs 96 men to build a house in 14 days. If he is asked to complete the building in 12 days, how many more men must he hire, assuming that the men work at the same rate?
- (ii) If y is inversely proportional to x , what is the percentage decrease in y when the value of x is tripled?

8. The diagram below shows the graph of the equation $y = (x + p)(3 - x)$ where p is a constant. The graph meets the y -axis at $A(0, 12)$ and the x -axis at B and C .



Find

- (i) the coordinates of point B ,
- (ii) the maximum point of the graph $y = (x + p)(3 - x)$.
- (iii) A point $(w, 6)$ lies on the curve. What are the possible values of w ?